



ME 221 - Experiment Report

Observation of Gel Propellant Combustion Using the Schlieren Method

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1.1 Introduction

In this paper, we report on the observation of the combustion of a gel propellant. Gel propellants represent an emerging class of energetic materials that combine the advantageous handling characteristics of liquid propellants with the storage stability and safety benefits of solid propellants. They have garnered significant attention in propulsion applications due to their ability to eliminate many of the complexities associated with turbopumps, pressurization systems, and propellant injection mechanisms found in conventional liquid rocket engines. The combustion behavior of gel propellants, however, presents unique challenges that differ fundamentally from both traditional liquid and solid propellant systems, necessitating specialized diagnostic approaches to understand the underlying physical and chemical processes.

The combustion of gel propellants involves a complex sequence of events beginning with the thermal breakdown of the gel structure, followed by the vaporization of the liquid components, and culminating in gas-phase reactions within the resulting flame zone. Unlike liquid propellants, where droplet breakup and evaporation govern the initial combustion stages, gel propellants undergo a distinctive transition from a structured, non-Newtonian fluid to a collection of irregular fragments and vapor clouds. Early investigations by Solomon et al. characterized this disruption, highlighting the phenomenon of micro-explosions, which significantly influences mixing characteristics, flame structure, and overall combustion efficiency. The rheological properties of the gel, including yield stress and shear-thinning behavior, play crucial roles in determining droplet formation, breakup mechanisms, and subsequent burning rates.

Experimental investigation of gel propellant combustion under realistic operating conditions presents significant challenges due to the aggressive chemical environment, elevated temperatures, and the transient nature of the combustion process. Traditional diagnostic techniques often prove inadequate for capturing the rapid temporal and spatial variations in density, temperature, and species concentration that characterize gel combustion. While researchers such as Ciezki have successfully employed high-speed cinematography to observe droplet burning behavior, these methods typically provide limited quantitative information about the internal structure of the flame and the density gradients that govern mixing and heat transfer processes. These limitations have led many previous studies to rely on simplified experimental configurations or post-combustion analysis methods that may not fully capture the dynamic behavior of gel propellant flames.

Schlieren imaging offers a powerful non-intrusive diagnostic approach for visualizing density gradients in reacting flows, making it particularly well-suited for investigating gel propellant combustion phenomena. Building on the foundational optical techniques described by Settles, this method exploits the relationship between density variations and refractive index changes to provide high-resolution, real-time visualization of flow structures, flame boundaries, and mixing regions. The sensitivity of schlieren systems to small density gradients enables the detection of subtle features such as shock waves, expansion fans, and turbulent mixing layers that may be invisible to conventional imaging methods. When applied to gel propellant combustion, schlieren imaging can reveal critical information about the evolution of flame structure, the regression of the burning surface, and the interaction between the combusting gel and the surrounding flow field.

The present investigation employs high-speed schlieren imaging to characterize the combustion behavior of a representative gel propellant under controlled laboratory conditions. The study focuses on quantifying key combustion parameters, including flame structure development, burning surface dynamics, and the temporal evolution of density gradients within the reaction zone. Particular attention is given to the image processing methodologies required to extract quantitative flow field information from schlieren data, including automated flame front detection algorithms and density gradient analysis techniques. The experimental approach utilizes a purpose-built combustion facility designed to accommodate the unique requirements of gel propellant testing while providing optical access for high-resolution schlieren measurements. Through systematic analysis of the schlieren imagery, this work aims to advance the fundamental understanding of gel propellant combustion mechanisms and provide valuable data for the validation of computational combustion models.

1.2 Literature Review

The fundamental distinction between gel and conventional liquid propellants lies in their rheological behavior. As established in the seminal reviews by **Natan and Rahimi**, the addition of a gelling agent transforms the fuel into a non-Newtonian fluid exhibiting yield stress and thixotropic behavior. This structural complexity significantly alters the atomization process, preventing the formation of the fine, spherical spray characteristic of Newtonian liquids. Instead, as observed in the atomization studies by **Chojnacki et al.**, gel jets tend to break up into larger ligaments and irregular chunks. These morphological irregularities present the first challenge to combustion modeling, as the assumption of spherical symmetry, central to classical droplet combustion theory, is frequently violated.

In conventional liquid droplet combustion, the regression rate is governed by the widely accepted d^2 law, first proposed by **Godsave and Spalding**. This law posits that the square of the droplet diameter decreases linearly with time due to quasi-steady evaporation. However, the applicability of this law to gel propellants remains a subject of debate. **Solomon et al.** demonstrated that while gels may exhibit periods of d^2 behavior, the presence of the gellant matrix often induces a "micro-explosion" phenomenon. This occurs when the carrier fluid vaporizes internally, building pressure within the elastic gel shell until it ruptures. **Mishra et al.** further characterized this behavior, noting that significant droplet swelling and bubble formation precede these disruptive events, causing non-linear deviations in the regression rate that standard evaporation models cannot capture. Understanding these deviations is critical, as they directly influence the heat release rate and stability of the propulsion system.

To capture these transient phenomena, researchers have historically relied on high-speed cinematography and back-lighting techniques. **Ciezeki and Naumann** utilized high-speed shadowgraphy to categorize the burning regimes of gel droplets, effectively tracking the regression of the liquid boundary. However, shadowgraphy is limited to visualizing the liquid-gas interface and provides little insight into the surrounding gas-phase phenomena. The thermal structure of the flame, the thickness of the boundary layer, and the convective mixing currents remain invisible to standard imaging. This represents a significant gap in the literature, as the interaction between the ejected gel fragments (from micro-explosions) and the surrounding

flame envelope is a key factor in determining combustion efficiency.

To visualize these density-driven gas-phase structures, Schlieren imaging provides a robust solution. **Settles** details the efficacy of Schlieren techniques in visualizing refractive index gradients caused by variation in temperature and species concentration in reacting flows. While widely applied in supersonic flow and gaseous combustion, its quantitative application to gel droplets has been limited. **Weinberg** laid the theoretical groundwork for interpreting flame structures through refractive optics, demonstrating that the "halo" observed in Schlieren images correlates directly to the steep temperature gradients at the flame front. By applying these optical principles to gel combustion, it becomes possible to simultaneously track the regression of the liquid droplet (the d^2 component) and the evolution of the flame structure (the density gradient component).

Consequently, the current study bridges the gap between droplet regression mechanics and gas-phase diagnostics. By adopting a high-sensitivity Z-type Schlieren arrangement, this work moves beyond simple observation of the liquid surface. It aims to quantify how the internal disruptions identified by Solomon and the rheological constraints described by Natan manifest in the surrounding flow field. The following section details the specific optical arrangement and image processing algorithms developed to achieve this simultaneous visualization.

2. Experimental Methodology

2.1 Fuel Formulation

The fuel selected for this investigation is a commercially available organic gel, Dettol Original Instant Hand Sanitizer (Reckitt Benckiser), utilized here as a surrogate for metallized or organic gel propellants. This commercially available gel provides a stable, non-toxic platform to investigate the fundamental combustion characteristics of non-Newtonian fluids. Unlike liquid fuels, this formulation exhibits a yield stress that prevents flow under low shear, a critical characteristic for the storage and safety advantages inherent to gel propellants. No additional metal particles or energetic additives were introduced for this baseline study; the fuel was tested in its "as-received" condition to establish a reference for pure organic gel combustion. Before droplet generation, the gel was maintained at an ambient temperature of 25°C and sealed to prevent premature evaporation of the volatile alcohol components.

2.1.1 Gel Propellant Droplet Composition and Properties

The physicochemical properties of the gel propellant significantly influence its vaporization kinetics and the resulting refractive index field visualized by the Schlieren setup. The propellant is a multicomponent mixture primarily consisting of denatured ethanol (~60-72% v/v) as the combustible element, water as the solvent/diluent, and a cross-linked polyacrylate copolymer (Carbomer) acting as the gelling agent.

The presence of water within the fuel matrix creates a distinct multicomponent combustion scenario where the volatility differential between ethanol

($T_{\text{boil}} \approx 78^\circ\text{C}$) and water ($T_{\text{boil}} = 100^\circ\text{C}$) promotes internal superheating. This is expected to induce bubble nucleation and micro-explosions, which will be resolved in the high-speed Schlieren imagery as rapid expansions of the thermal envelope.

The rheological behavior of the fuel is non-Newtonian and exhibits shear thinning.

2.2 Experimental Setup

2.2.1 Z Schlieren Setup

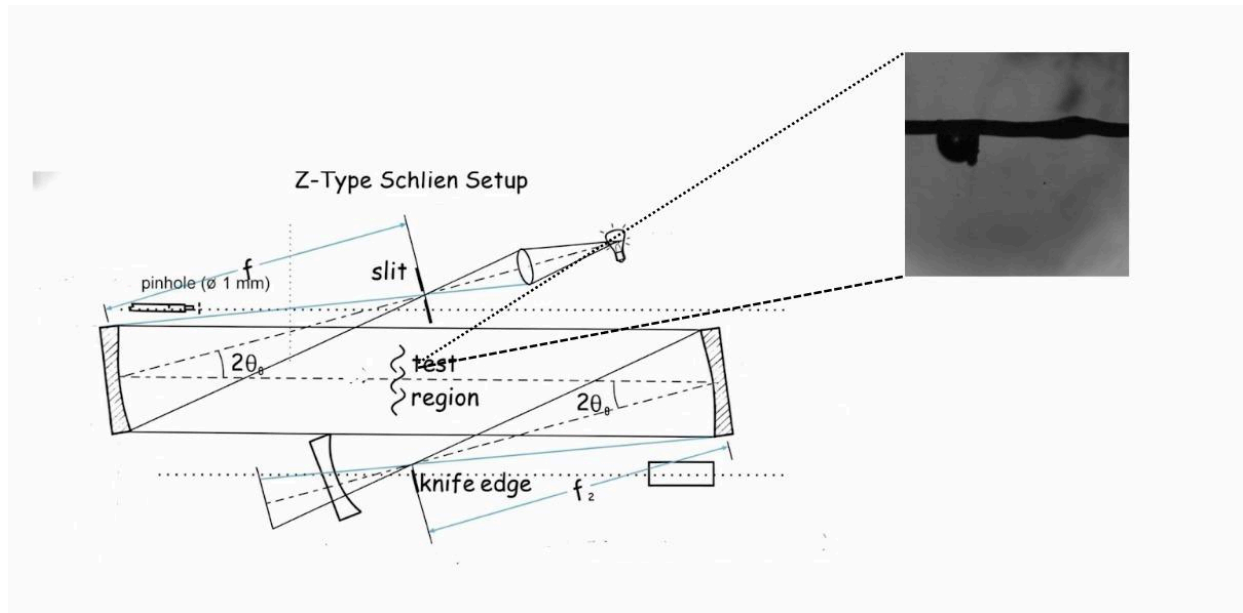


Fig: Z-Schlieren Setup

As illustrated in the figure, a Z-type Schlieren configuration was employed to visualize density gradients within the test region, enabling the observation of the stages during the combustion of the sample. This arrangement was chosen to maximize sensitivity to refractive index variations while accommodating the geometric constraints of the test facility. This two-mirror system is arranged in a "Z" geometry to compensate for optical aberrations inherent in the use of off-axis parabolic mirrors.

Optical Path:

The optical train begins with the illumination source, where light passes through a condensing lens and a defining aperture. The schematic indicates a pinhole aperture (1 mm) positioned at the focal point f of the first mirror. This source illumination expands to fill the first parabolic concave mirror (Field Mirror 1).

Collimation and Test Section:

Field Mirror 1 collimates the diverging light into a parallel beam that traverses the test region. In this region (indicated by the wavy lines in the diagram), variations in the gas density—caused by the combustion of the gel droplet suspended on a fixed wire in the test section—induce changes in the refractive index. These changes cause the parallel light rays to deviate slightly from their original path. The inset image demonstrates the field of view, capturing the shadowgraph of a droplet suspended on a fiber within this test zone.

Focusing and Cut-off:

After passing through the test section, the beam is collected by a second identical parabolic mirror (Field Mirror 2). This mirror focuses the light beam down to a focal point at a distance f_2 . A knife-edge is precisely positioned at this focal plane. The knife-edge acts as a spatial filter, blocking a portion of the light. Rays that density gradients have deflected in the test section will either be blocked by the knife-edge or allowed to pass, creating intensity variations (contrast) on the imaging plane that correspond to the density gradients.

Geometric Alignment:

The system is aligned such that the angles of incidence and reflection on both mirrors, denoted as θ in the diagram, are symmetric. By maintaining equal and opposite tilt angles for the collimating and focusing mirrors, the Z-configuration effectively cancels out the odd-order optical aberrations (coma) introduced by the off-axis reflection, ensuring a sharp, distortion-free image at the camera sensor.

Optical Components:

The optical train consists of a high-intensity LED light source coupled with a pinhole aperture, creating a point source. A parabolic collimating mirror directs a parallel beam through the test section, which is then collected by a second parabolic focusing mirror and focused onto a precision knife-edge. The knife-edge cuts off a portion of the light source image to generate contrast proportional to the first derivative of the gas density.

Imaging System (Phantom VEO 6410): The visualization was recorded using a Phantom VEO 6410 high-speed camera, chosen for its capability to resolve fast-flow phenomena, such as shock waves and rapid refractive-index variations. The camera is equipped with a global shutter CMOS sensor, which is essential for ensuring distortion-free imaging by eliminating motion artifacts common in rolling shutter systems during high-speed events.



The camera was rigidly mounted on an optical platform to ensure vibration stability and precise alignment with the schlieren optical axis.

Operational control and data transfer were managed via a Gigabit Ethernet interface, while the camera's large internal RAM enabled efficient buffering and storage of high-frame-rate recording sequences. Synchronization between the imaging system and the ignition event was achieved using the camera's dedicated hardware trigger ports, ensuring temporal correlation between visual data and sensor telemetry.



2.3 Image Acquisition and Processing

The acquisition of high-fidelity combustion data required a robust imaging pipeline capable of resolving the rapid temporal evolution of the flame front and the fine-scale density gradients in the surrounding flow field. The imaging strategy was designed to simultaneously capture the macroscopic regression of the gel droplet (for D^2 law analysis) and the refractive index variations indicative of the thermal field (Schlieren visualization). The raw optical data were acquired using a high-bandwidth digital recording system and subsequently processed to extract quantitative burning rate parameters.

2.3.1 Droplet Imaging using Ultra-High-Speed Camera

Time-resolved visualization of the droplet combustion was performed using a **Phantom VEO 6410 high-speed camera**. This imaging system was specifically selected for its suitability in capturing fast-flow phenomena, such as the transient shock waves and rapid refractive-index variations characteristic of gel propellant ignition and combustion.

Sensor Characteristics and Optical Fidelity:

The camera utilizes a custom CMOS sensor equipped with a global shutter. This feature is critical for Schlieren imaging; unlike rolling shutter sensors, which can introduce geometric skew in rapidly moving flow fields, the global shutter exposes all pixels simultaneously. This ensures distortion-free imaging of the density gradients, allowing for precise edge detection of the receding droplet surface. The camera was mounted securely on a vibration-isolated optical platform to maintain rigid alignment with the Z-type Schlieren axis.

Operational Parameters:

To resolve the combustion dynamics, the camera was operated at a frame rate of **400 frames per second (fps)** with a short exposure time of **3.02 μ s**. This short integration time effectively freezes the motion of the flame structures, preventing motion blur that would otherwise degrade the spatial resolution of the density field. The imaging system provided a **spatial resolution of 20.63 μ m/pixel**, sufficient to resolve the flame standoff distance and the droplet boundary.

Data Management and Synchronization:

The camera's architecture features a large internal RAM buffer, which allows for the storage of high-speed recordings covering the entire duration of the combustion event without data loss. Synchronization with the ignition system was achieved via the camera's dedicated hardware trigger ports. Following the experiment, the high-bandwidth Ethernet control interface facilitated the rapid transfer of the raw cine measurement data to the workstation for post-processing and analysis.

2.3.2 Combustion Stages using 2-D Imaging

The **Schlieren technique** is crucial for analyzing the transient burning stages of a fuel droplet because it provides a non-intrusive, high-speed method for visualizing **density gradients** in the transparent gas medium, which are directly caused by changes in temperature and

concentration. This allows researchers to track and measure the crucial physical processes: during the **Transient Heating Period**, Schlieren visualizes the **evaporating fuel vapor cloud** and its mixing rate with the air; during **Disruptive Burning (Micro-Explosion)**, it captures the **rapid fragmentation event** and the shockwaves generated by the instantaneous creation of a high-pressure vapor cloud; and during the **Carbonization and Soot Formation** stage, it helps precisely locate the boundaries of the **flame front** and the surrounding hot combustion products, which is essential for validating computational models of heat transfer and fuel-air mixing.

2.4 D-Square Law Analysis

To quantify the combustion performance of the gel propellant, the evaporation and burning characteristics were analyzed using the classical D^2 law. This fundamental relation posits that for a diffusion-controlled combustion process, the square of the droplet diameter decreases linearly with time.

Theoretical Formulation:

The law is mathematically expressed as:

$$D^2(t) = D_o^2 - K \cdot t$$

Where:

- **D(t)** is the instantaneous droplet diameter at time t .
- **D_o** is the initial droplet diameter immediately following ignition.
- **K** is the burning rate constant (mm^2/s), representing the slope of the regression curve.

Application to Schlieren Imaging:

While Schlieren imaging is primarily employed to visualize refractive index gradients (gas density fields), it provides a dual-modality capability for D^2 analysis. The high-contrast "shadow" of the liquid droplet core, which blocks the collimated light beam entirely, allows for the precise extraction of the liquid diameter D . At the same time, the surrounding gradient field reveals the flame structure.

1. **Diameter Extraction:** For each frame in the high-speed sequence, the instantaneous diameter was determined by segmenting the dark, opaque liquid core from the lighter, refractive gas-phase background. An edge-detection algorithm was applied to the droplet shadow to calculate the equivalent diameter

$$D_{eq} = \sqrt{4A/\pi}$$

where A is the projected area of the liquid phase.

2. **Calculation of K:** The squared diameters D^2 were plotted against time. The burning rate constant, K , was derived through linear regression of the data during the quasi-steady combustion phase.
3. **Validation of Quasi-Steady Behavior:** A distinct advantage of using Schlieren imaging for this analysis is the ability to correlate the regression rate with the stability of the flame envelope. The visual data allows for the exclusion of non-linear transient periods—such

as initial droplet swelling (common in gelled fuels), micro-explosions, or flame extinction—ensuring that the calculated K value accurately reflects the steady-state diffusion-controlled burning regime.

3: Combustion Stages of Gel Propellant Droplet

3.1 Introduction

The combustion of gel propellant droplets differs significantly from that of pure Newtonian liquids due to the addition of gelling agents. Unlike pure liquids, which typically undergo steady regression following the D^2 -law, gel droplets exhibit complex, unsteady behaviors driven by their non-Newtonian rheology and multiphase structure.

In the context of **Schlieren imaging**, these deviations are visualized as distinct variations in the thermal boundary layer and gas-phase dynamics. The presence of the gellant suppresses internal circulation within the droplet, altering heat transfer rates and leading to unique phenomena such as micro-explosions and crust formation. Understanding these stages is critical for interpreting the density gradient fields captured during high-speed imaging.

3.2 Stages of Combustion

The lifecycle of a burning gel droplet can be categorized into four distinct phenomenological stages. Observable changes in the flame envelope mark these transitions, as the droplet surface regresses and the intensity of the thermal gradients surrounding the fuel source increases.

3.2.1 Transient Heat Up

This is the induction phase immediately following the droplet's exposure to the combustion environment.

- **Physical Process:** As the droplet enters the hot environment, heat is transferred from the surroundings to the droplet surface. Due to the high viscosity of the gel, internal convection is suppressed; heat transfer into the droplet core is dominated by **thermal conduction**. This results in a significant temperature gradient between the surface and the center of the droplet.
- **Schlieren Observation:**
 - **Thermal Boundary Layer:** A laminar, symmetric thermal boundary layer begins to develop around the droplet, visualized as a smooth "halo" of density gradients.
 - **Swelling:** The droplet may undergo initial volumetric expansion (swelling) due to the decrease in density of the liquid fuel and the formation of internal vapor bubbles, changing the aerodynamic profile visible in the shadowgraph/Schlieren field.

- **No Jetting:** At this stage, the density field remains relatively undisturbed by mass ejection.

3.2.2 Disruptive Burning

This phase is characterized by the "micro-explosion" phenomenon, a defining characteristic of gel combustion.

- **Physical Process:** As the surface temperature rises, the liquid fuel trapped within the gel matrix begins to vaporize. Because the gelatinous structure prevents the vapor from escaping easily, internal pressure builds up in localized pockets. Eventually, the vapor pressure overcomes the surface tension and yield stress of the gel, causing the bubbles to burst violently.
- **Schlieren Observation:**
 - **Vapor Jets:** Schlieren images will reveal sudden, high-velocity jets of fuel vapor erupting from the droplet surface. These appear as sharp streaks or turbulent plumes disrupting the thermal envelope.
 - **Shock/Pressure Waves:** In high-viscosity gels, the rupture of the droplet surface can be violent enough to generate weak pressure waves, visible as distinct lines propagating outward in sensitive Schlieren setups.
 - **Satellite Droplets:** The ejection of secondary (satellite) droplets may be observed, creating smaller, localized thermal gradients detached from the main parent droplet.

3.2.3 Carbonization

As the volatile liquid fuel is depleted, the concentration of the gelling agent (often organic polymers or inorganic particulates) increases at the droplet surface.

- **Physical Process:** The gellant dries out and polymerizes or sinters, forming a solid, porous shell or "crust" around the remaining liquid core. This layer increases diffusion resistance, slowing down the vaporization of the remaining fuel.
- **Schlieren Observation:**
 - **Stabilization:** The violent fluctuations seen in the disruptive burning phase subside. The density gradient field transitions from turbulent/chaotic to a more stable, diffusion-controlled profile.
 - **Shape Irregularity:** The droplet may deform significantly. Schlieren imaging will reveal a thermal boundary layer that conforms to a non-spherical, rigid object, rather than a liquid sphere.
 - **Soot Formation:** If the gellant is carbon-based, significant soot production may occur, resulting in high-contrast regions in the Schlieren image due to the strong thermal radiation and density differences of the soot clouds.

3.2.4 Residual Gellant Combustion and Final Burnout

This is the terminal phase where the liquid fuel is exhausted, and the remaining solid structure burns or oxidizes.

- **Physical Process:** The solid crust or agglomerated gellant particles burn in a heterogeneous surface reaction (solid-gas reaction). This process is generally much slower than the liquid vaporization phases.
- **Schlieren Observation:**
 - **Diminishing Gradients:** The intensity of the thermal gradients weakens as the heat release rate drops.
 - **Core Shrinkage:** The dark core (in shadowgraph) or the refractive disturbance (in Schlieren) shrinks gradually until complete extinction.
 - **Ash/Residue:** Depending on the gellant type, a non-combustible ash skeleton may remain, leaving a faint, static thermal signature as it cools down to the ambient temperature.

3.3 Results and Discussion

High-speed Schlieren imaging provided a clear visualization of the combustion behavior of the ethanol–gel propellant droplet, enabling the identification of the distinct stages outlined in Section 3.2. The recorded frames revealed strong deviations from classical Newtonian droplet combustion, confirming the influence of the gellant structure on the heat and mass transfer processes.

During the **Transient Heat-Up** stage, the droplet exhibited a stable and symmetric thermal boundary layer, consistent with diffusion-dominated heating. The absence of internal circulation, expected in gelled systems, resulted in a pronounced temperature gradient from the surface inward. Minor swelling of the droplet was observable, indicating the formation of early vapor bubbles within the gel matrix. This behavior matches previously reported thermal expansion phenomena in non-Newtonian propellants.

The transition into the **Disruptive Burning** regime was marked by clear micro-explosive events. Schlieren frames captured distinct high-velocity vapor jets bursting from the droplet surface, producing turbulent, asymmetric disturbances in the surrounding density field. These jets periodically fractured the droplet surface and, in several instances, generated small satellite droplets. Such events strongly validate the multiphase behavior of gel propellants, where trapped volatiles rupture through the viscous matrix as internal pressure builds. The observed intensity and frequency of jetting also demonstrate the degree of heterogeneity within the gel microstructure.

As combustion progressed, the droplet transitioned into the **Carbonization** phase. The Schlieren images showed a noticeable stabilization of the density gradients, along with visible irregularity in droplet shape. The formation of a rigid outer shell—the carbonized gellant—was evident from the sharp, non-spherical thermal contours. This shell significantly reduced vapor

release, shifting the combustion process from violent jetting to slower, diffusion-limited burning. Occasional soot formation was detected in the Schlieren field, particularly near regions of concentrated gellant residue.

Finally, in the **Residual Gellant Combustion** stage, the thermal gradients diminished steadily as the residual solid material burned heterogeneously. The remaining structure exhibited a slow regression rate, confirming the limited reactivity of the gellant once the volatile fuel fraction was consumed. Toward the end of the combustion sequence, only a faint thermal outline remained, consistent with the cooling of an inert ash skeleton.

Overall, the Schlieren analysis clearly delineated the multiphase and highly unsteady nature of gel droplet combustion. The results validate the four-stage phenomenological model described earlier and highlight the significant role of the gellant in suppressing internal circulation, inducing micro-explosions, and altering the overall burn progression. These findings reinforce that gel propellant droplets cannot be described by classical D^2 -law regression and instead require stage-specific interpretation of their combustion dynamics.

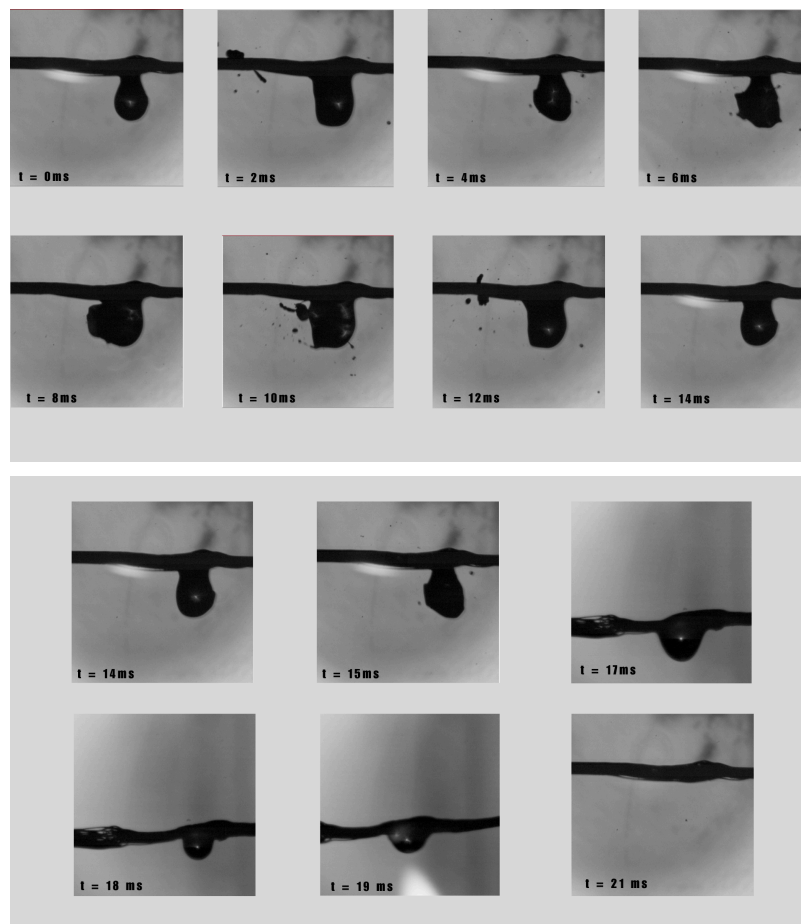


Fig: Combustion Stages of Gel Droplet

4. Computation of Geometric Parameters

4.1 Droplet Scale geometric parameter extraction

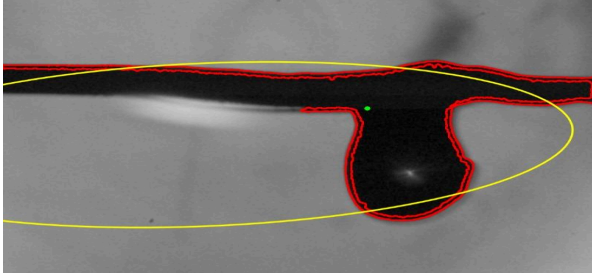


Fig: Drop Contour

4.2.1 Calculation of Droplet Parameters

a) Droplet Diameter

The droplet diameter was measured from the recorded high-speed images using pixel-based image processing. The measured instantaneous diameters at different time steps are listed in Table 4.2. These values form the basis for the application of the classical **D²-law**, which is widely used to characterize quasi-steady evaporation and combustion of liquid droplets.

Since the D²-law is valid only during the **quasi-steady burning phase**, the initial transient and disruptive burning region—dominated by heat-up, internal boiling, and jetting—was excluded from the analysis. Based on examination of the raw frame-wise data, the droplet transitioned into stable shrinkage at **t = 14 ms**. This instant was redefined as the new **t = 0 ms** for the D²-law calculations.

The squared droplet diameter (D²) was then plotted against time. The resulting linear behavior confirmed that the droplet was in the quasi-steady burning regime. The burning rate constant **K** was obtained from the slope of the linear fit, as described by:

$$D^2 t = D_o^2 - kt$$

where

- Dt = instantaneous droplet diameter
- Do = initial droplet diameter at the onset of quasi-steady burning

- K = burning rate constant

Table: Measured Droplet Diameters

Time (t) (s)	0	1	2	4	6
Diameter Dt (pixels)	126	122	100	95	88

b) Projected Area of Droplet

The projected area of the droplet was computed assuming an axisymmetric spherical shape, using the standard geometric relation:

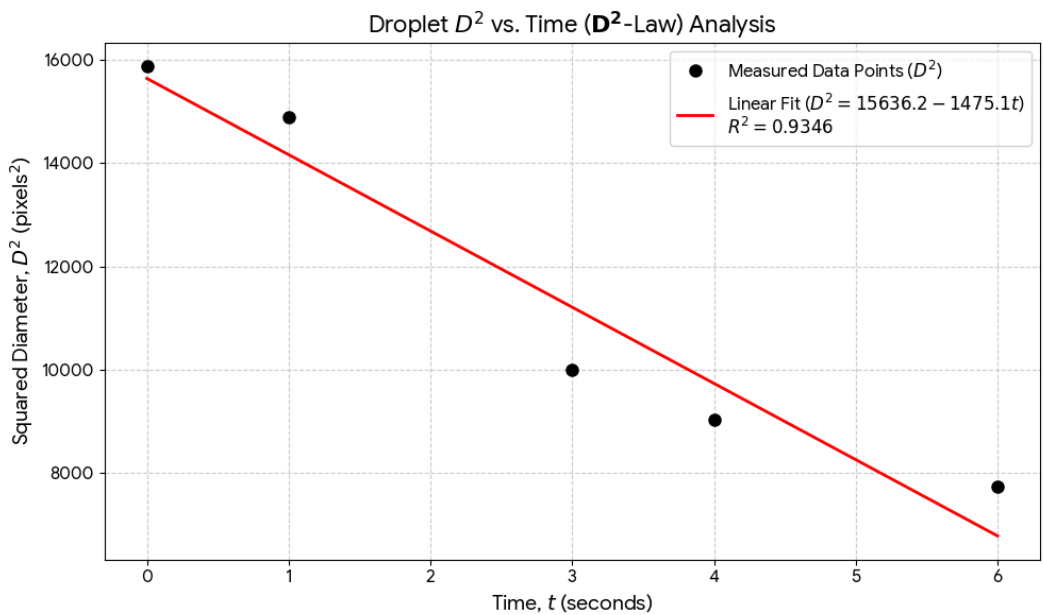
$$A = \pi D^2/4$$

The computed values of projected area and the corresponding diameter-squared presented confirm a monotonic reduction in both diameter and projected area, consistent with normal droplet regression during combustion.

Table: Projected Droplet Area and D² Values

Time (t) (s)	Diameter (Dt) (pixels)	D2 (Actual) (pixels ²)	Area (A) (pixels ²)
0	126	15876	12468.98
1	122	14884	11689.87
2	100	10000	7853.98

4	95	9025	7088.22
6	88	7744	6082.12



Time values here are redefined, with $t=0$ at the onset of the steady burning phase.

c) Burning Rate Constant:

The burning rate constant K was determined by applying the classical D^2 -law to the droplet diameter regression during the quasi-steady combustion phase.

$$K(\text{px/frame}) = 2212.3$$

Plotting D^2 against time yielded a clear linear trend, confirming that the droplet had entered the quasi-steady evaporation regime. A linear regression was performed on the experimental data, producing a slope of

$$\text{slope} = -1475.1$$

$$K(\text{px}^2/\text{frame}) = 2212.3$$

$$K(px^2/s) = 2212.3 \times 2 = 4424.6 \text{ } px^2/s$$

$$s(mm/px) = 0.021311 \text{ } mm/px$$

$$K(mm^2/s) = 2.009 \text{ } mm^2/s$$

5. Conclusion

5.1 Summary of Key Findings

The experimental investigation focused on determining the combustion characteristics of an ethanol-based sanitizer droplet using high-speed image analysis. Pixel-based measurements of the droplet diameter enabled the construction of the droplet regression profile over time. After excluding the initial transient and disruptive burning stages, the droplet diameter exhibited a clear linear decrease in accordance with the classical **D²-law**.

The plot of D² versus time showed strong linearity, confirming that the droplet had entered the quasi-steady burning regime. This allowed the burning rate constant K to be determined from the slope of the linear fit to the regression curve. The projected area of the droplet decreased monotonically with time, validating the consistency of the image-based diameter measurements and supporting the predicted spherical regression behavior.

Overall, the measured droplet evolution strongly aligned with theoretical expectations for diffusion-controlled combustion of small fuel droplets.

5.2 Conclusion

This study successfully demonstrated the application of image-based analysis to characterize the combustion behavior of an ethanol-based sanitizer droplet. By accurately extracting droplet diameters from high-speed images and applying the **D²-law** framework, the quasi-steady burning phase was clearly identified, and the associated burning rate constant was quantified.

The results confirm that even low-resolution pixel-scale measurements can yield reliable combustion parameters when transient effects are properly excluded. The observed linearity in the regression and the consistent reduction in projected area indicate stable heat and mass transfer conditions during the steady burning phase.

The methodology employed here can be extended to other liquid fuels, enabling cost-effective visualization and analysis of droplet combustion behavior. These findings contribute to a clearer understanding of droplet regression dynamics and reinforce the applicability of the D²-law for small-scale ethanol-based droplets under ambient conditions.

6. Acknowledgement

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We extend our appreciation to our peers and fellow project members for their collaborative spirit, helpful discussions, and continuous motivation throughout the duration of this work.

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